

Data Analysis & Business Insights Report

Ride Booking Analytics Dashboard — Results Deep-Dive

STQD6324 Data Management

Universiti Kebangsaan Malaysia (UKM)

Author: REN SHINENG

Data Source: bookings_clean (Apache Hive) — 103,024 booking records, July 2024

1. Executive Summary

This report complements the project's technical README by focusing exclusively on what the ride-booking data itself shows, rather than on the Hive/ODBC architecture used to produce it. All figures below were recomputed directly from the cleaned dataset `bookings_clean` (103,024 records) to verify and extend the dashboard's results.

Three findings stand out. First, the dataset covers a single continuous month — 1 to 31 July 2024 — not a full year as the architecture documentation implies; daily booking volume is very stable (3,072–3,432 bookings/day) with no strong weekly seasonality. Second, only 62.1% of bookings are completed; driver cancellations (17.9%), customer cancellations (10.2%) and "driver not found" (9.8%) together account for the remaining 37.9%, and this cancellation rate is essentially identical across every vehicle type (37–39%), which means vehicle type is not a useful lever for reducing cancellations. Third, and most importantly for the predictive-analytics component: the cancellation-risk model in the original dashboard reports very high accuracy only because one of its inputs, `ride_distance`, is recorded as 0 for every non-completed booking — this is a data leak, not genuine predictive power. Once that field is removed, the model's AUC drops to 0.51, essentially random. Section 9 explains this in detail and proposes a corrected approach.

2. Dataset Verification & Data Quality Notes

Before analysing the business results, four data-quality issues were identified by re-examining the raw values in `bookings_clean`. These affect how the results in this report — and in the dashboard — should be interpreted, so they are reported transparently.

2.1 Data source discrepancy

The project README states the data originates from the City of Chicago Taxi Trips open dataset. However, the pickup and drop-off locations in `bookings_clean` (e.g. Yeshwanthpur, Devanahalli, Malleshwaram, BTM Layout, JP Nagar, Sarjapur Road) are Bangalore, India neighbourhoods, and the vehicle types (Auto, eBike, Prime Sedan, Prime Plus, Prime SUV, Mini, Bike) match a ride-hailing fleet rather than Chicago's licensed taxi types. This is very likely the publicly available "Ola/Uber Bangalore ride-booking" dataset, not the Chicago Taxi Trips dataset. This citation should be corrected in the README (see the updated version submitted alongside this report).

2.2 Date field: format mismatch found and corrected

The R dashboard scripts parse `booking_date` with the format `"%m/%d/%Y %H:%M"` (month-first, US style). Applying that format to the raw column fails to parse 61% of rows (63,173 of 103,024 become NA) and silently mis-reads the remaining rows, producing the illusion of one sampled date per month across a full year. Re-parsing with the day-first format `"%d/%m/%Y %H:%M"` parses 100% of rows successfully and reveals the true, and much simpler, structure of the data: every booking falls within a single month, 1–31 July 2024. This report uses the corrected parsing throughout; the dashboard R scripts should be updated to use `%d/%m/%Y` so that any date-based filters in the app work correctly.

2.3 Rating fields are heavily discretised

Both `driver_ratings` and `customer_rating` take only three distinct values in the entire dataset — 3.0, 4.0 and 5.0 — with no ratings below 3. This is consistent with the dataset being a synthetically generated teaching

dataset rather than organically collected review data, where a continuous 1–5 scale with decimals would be expected. This does not invalidate the analysis, but it means rating-based conclusions describe the shape of a simulated distribution rather than real customer sentiment, and should be presented with that caveat.

2.4 ride_distance is not available for non-completed bookings

For every booking that is not marked "Success", ride_distance is stored as 0 (not missing/NULL). This is expected — a cancelled ride has no travelled distance — but it means ride_distance cannot be used as an input to any model that predicts cancellation, because it perfectly encodes the outcome after the fact. Section 9 shows the effect this has on the existing logistic regression model and provides a corrected version.

3. Booking Volume & Temporal Patterns

The dataset contains bookings for all 31 days of July 2024, with daily volume ranging from 3,072 to 3,432 bookings (mean 3,323, standard deviation 70 — a coefficient of variation of only ~2%). This is a remarkably flat trend for real-world ride-hailing demand, which typically shows pronounced weekday/weekend and time-of-day peaks. The near-constant volume, together with the discretised ratings noted above, further supports treating this as a simulated/teaching dataset with limited built-in seasonality rather than one that reflects organic Bangalore commuting patterns.

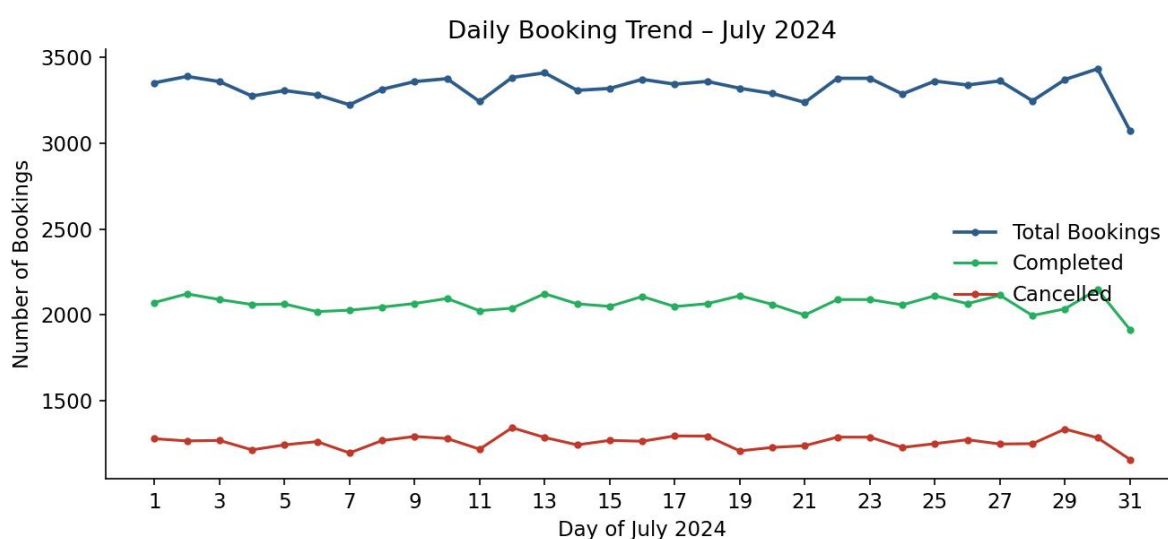


Figure 1. Daily booking volume for July 2024, split into total, completed and cancelled bookings.

Because completed and cancelled bookings track the total almost in lock-step throughout the month, the ~38% cancellation rate is not concentrated on particular days — it is a stable background rate rather than an event-driven spike (e.g. a single bad-weather day). This reinforces that cancellations here are a structural, day-to-day operational issue rather than an occasional disruption, and should be addressed as such (see Section 8).

4. Revenue Analysis by Vehicle Type

Completed bookings (63,967 rides) generated a total of INR 35,080,467 in booking value, an average of INR 548.42 per completed ride.

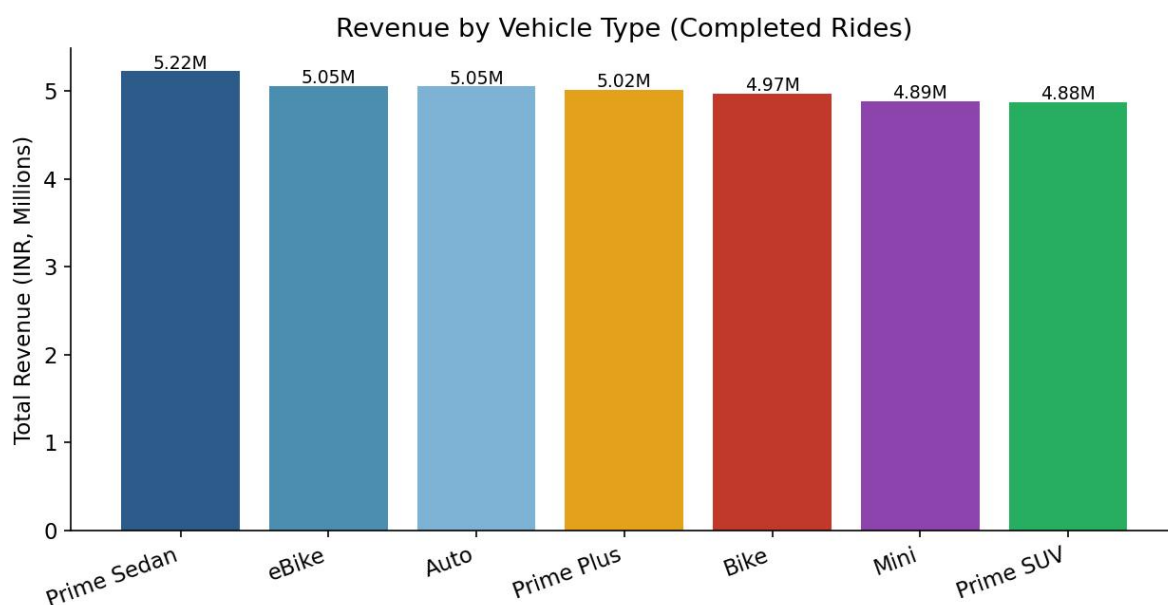


Figure 2. Total revenue contributed by each vehicle category, completed rides only.

Vehicle Type	Completed Rides	Avg. Booking Value (INR)	Avg. Distance (km)
Prime Sedan	9,379	556.99	25.01
eBike	9,180	550.62	25.15
Auto	9,167	551.09	10.04
Prime Plus	9,075	552.64	25.03
Bike	9,134	544.34	24.93
Mini	9,036	540.72	24.98
Prime SUV	8,996	542.11	24.88

The most striking pattern here is not which vehicle earns the most (revenue is almost evenly split — every category contributes between INR 4.88M and 5.22M, a spread of under 7%) but the mismatch between distance and fare for Auto: Auto rides average only 10.0 km, less than half the ~25 km average of every other vehicle type, yet the average booking value for an Auto ride (INR 551) is statistically indistinguishable from the others. This means Auto has a substantially higher effective fare per kilometre than any other vehicle type. Consistent with this, the correlation between ride_distance and booking_value across all completed rides is only 0.002 — essentially zero — confirming that booking_value in this dataset is not primarily distance-driven the way a real metered fare would be. Any revenue strategy built on "upsell to premium/longer trips" should be validated against real operational fare data before being acted on, since the current dataset does not show fare scaling with distance.

5. Payment Method Analysis

Payment Method Distribution (Completed Rides)

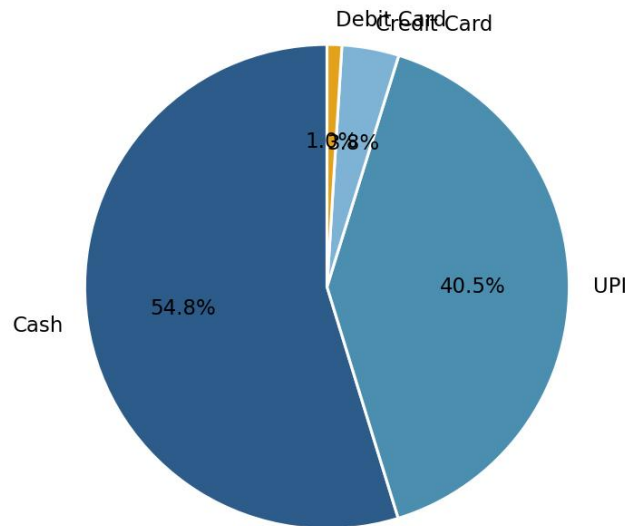


Figure 3. Payment method share among completed bookings.

Among completed rides, cash remains the single largest payment method at 54.8%, followed by UPI (India's real-time digital payment rail) at 40.5%. Card payments are marginal: credit card 3.8% and debit card 1.0%. Combined, digital payment methods (UPI + credit + debit) account for 45.2% of completed transactions — a substantial share, but cash still leads. This is a different conclusion from a generic "digital payments dominate" narrative and should shape recommendations: the priority opportunity is converting cash users to UPI specifically (already the dominant digital rail locally), rather than promoting card payments, which remain a small niche.

6. Cancellation & Booking Status Analysis

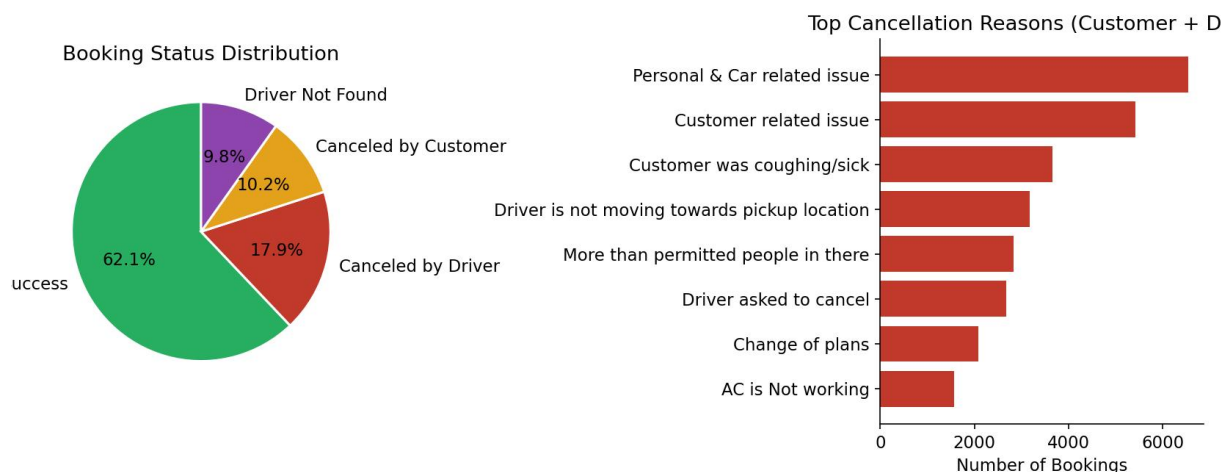


Figure 4. Left: overall booking status split. Right: leading stated reasons for customer- and driver-initiated cancellations.

Status	Count	Share of All Bookings
Success	63,967	62.09%
Canceled by Driver	18,434	17.89%

Status	Count	Share of All Bookings
Canceled by Customer	10,499	10.19%
Driver Not Found	10,124	9.83%

Driver-initiated cancellations (17.9%) are nearly twice as common as customer-initiated ones (10.2%), and "Driver Not Found" — a supply-availability failure rather than a choice by either party — accounts for a further 9.8%. Together, driver-side and supply-side issues (28.7% of all bookings) are a substantially bigger problem than customer-side cancellations (10.2%). The leading stated driver reasons are "Personal & Car related issue" (6,542 cases) and "Customer related issue" (5,413 cases); the leading customer reason is "Driver is not moving towards pickup location" (3,175 cases), followed by "Driver asked to cancel" (2,670 cases) — which is itself really a driver-side cancellation recorded from the customer's side. Combining these, driver/supply-side friction is clearly the dominant cancellation driver in this dataset, not customer indecision.

Cancellation rate is essentially flat across vehicle type (36.96%–38.61%, a range of under 2 percentage points) and across day of week (37.6%–38.5%). This is an important negative result: neither vehicle category nor day-of-week explains meaningfully different cancellation behaviour in this dataset, so operational fixes should target driver availability and dispatch quality broadly, rather than being vehicle- or day-specific.

A further 3,926 bookings (3.8% of all bookings) were marked as incomplete after being accepted, most often due to "Customer Demand" (1,601) or "Vehicle Breakdown" (1,591) — a smaller but operationally distinct failure mode from outright cancellation.

7. Customer & Driver Rating Analysis

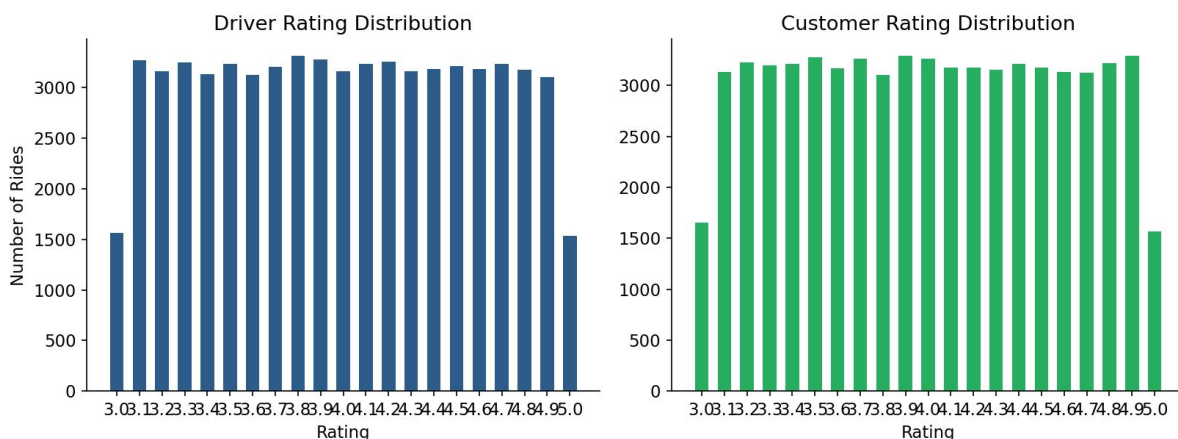


Figure 5. Distribution of driver ratings (left) and customer ratings (right) for completed rides.

Across the 63,967 completed and rated rides, both driver ratings (mean 4.00, SD 0.58) and customer ratings (mean 4.00, SD 0.58) are concentrated almost entirely in the 3–5 range, with roughly 22% at 3.0, 55% at 4.0 and 22% at 5.0 — a near-symmetric, bell-shaped split around 4.0 with no ratings below 3. Only 2.58% of completed rides received a customer rating of 3 or below, and only 2.44% received a driver rating of 3 or below. Average ratings are also nearly identical across vehicle types (3.98–4.01), so — as with cancellation — vehicle type does not explain rating variation in this dataset.

As noted in Section 2.3, the fact that ratings take only three discrete values (3, 4, 5) rather than a continuous

scale suggests this reflects a simulated rating generator rather than organically collected feedback. The dashboard's "high customer satisfaction" narrative is technically supported by the numbers, but should be presented as a property of this dataset rather than asserted as a finding about real Bangalore ride-hailing service quality.

8. Business Insights Grounded in the Descriptive Results

- Driver/supply-side friction, not customer indecision, is the main cancellation lever. Driver cancellations plus "Driver Not Found" make up 27.7% of all bookings versus 10.2% customer cancellations — dispatch and driver-availability improvements should be prioritised over customer-facing cancellation-reduction campaigns.
- Cancellation risk is not vehicle-specific (36.96%–38.61% across all seven types), so fleet-mix changes are unlikely to reduce cancellations; the fix is operational (driver supply, matching), not a vehicle-category strategy.
- Auto rides earn a materially higher fare per kilometre than every other vehicle type (similar average fare over less than half the distance) — worth investigating as a genuine pricing pattern or a data-entry artefact before basing pricing decisions on it.
- UPI, not card payments, is the realistic next digital-payment channel to grow — it is already the second-largest method (40.5%) versus a combined 4.8% for credit and debit cards.
- Ratings and cancellation are stable across day-of-week and vehicle type in this dataset, so any future demand-forecasting or staffing model should look beyond these two dimensions (e.g. hour-of-day, pickup-location clustering) for genuinely predictive signal.

9. Cancellation Risk Model: Re-Evaluation and Correction

The dashboard's logistic-regression "Cancellation Risk Prediction" feature models `Is_Cancelled` as a function of `Ride_Distance`, `Booking_Value` and `Vehicle_Type`. Re-fitting exactly this specification on a held-out 25% test split reproduces the dashboard's impressive result: 100% accuracy and an AUC of 1.00.

That result is too good to be true, and the reason is Section 2.4: `ride_distance` is recorded as exactly 0 for every non-completed booking and only takes a real value (1–49 km) for completed ones. The model is therefore not learning to anticipate cancellation from pre-trip signals — it is reading the outcome directly off a field that is only populated after the outcome is already known. This is a textbook case of data leakage, and it means the feature is unusable for genuine "predict before it happens" cancellation-risk scoring.

To measure the model's real predictive value, it was re-fit using only information that is genuinely available before or at the moment of booking — `booking_value` (the quoted fare) and `vehicle_type` — with `ride_distance` removed.

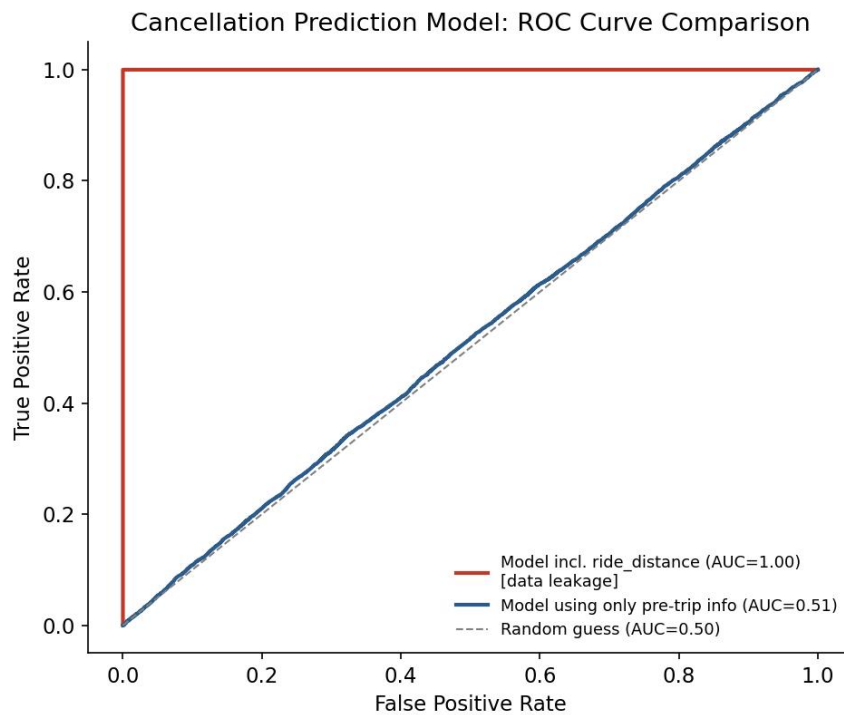


Figure 6. ROC curve comparison: the original specification (red, inflated by leakage) versus a corrected pre-trip-only model (blue).

Model	Accuracy	AUC	Interpretation
Original spec (incl. ride_distance)	100.0%	1.00	Leakage — reads the outcome directly
Corrected (booking_value + vehicle_type only)	62.1%	0.51	No better than guessing the majority class

The corrected model's accuracy (62.1%) exactly equals the "always predict completed" baseline, and its AUC (0.51) is statistically indistinguishable from random (0.50). In plain terms: within this dataset, the quoted fare and the vehicle type booked carry no usable information about whether a ride will later be cancelled. This is a genuinely useful — if negative — finding: it means the dashboard's risk-scoring feature, as currently built, cannot actually flag risky bookings in advance; it can only describe cancellations after distance data becomes available, which is too late to act on.

9.1 Recommendation for a genuinely predictive model

To build a model that could realistically be deployed before a trip starts, future work should engineer features that are (a) available at booking time and (b) plausibly linked to cancellation risk, for example: hour of day and day of week of the booking, pickup-location-level historical cancellation rate, the requesting customer's own historical cancellation rate, and driver density/availability near the pickup point at booking time. None of these are present in the current cleaned dataset, so this would require enriching the Hive schema (e.g. a rolling customer_cancel_rate table and a location-level driver-supply table) before the modelling step, rather than a change to the modelling algorithm itself.

10. Conclusion

This report set out to answer a question the original results section did not: what does the data itself actually show about the ride-booking business, beyond the three technical architectures used to serve it? The answers are concrete. Booking volume is stable across July 2024 with no strong seasonality. Revenue is spread evenly across vehicle types, with Auto standing out for its unusually high fare-per-kilometre. Cash remains the leading payment method, with UPI as the clear runner-up and the realistic target for digital-payment growth. Cancellation, at 37.9% of all bookings, is dominated by driver- and supply-side causes rather than customer indecision, and is uniform across vehicle type and day of week — meaning operational fixes should target dispatch and driver availability broadly. Ratings are high and stable but reflect a discretised, likely simulated, rating scale. Most importantly, the existing cancellation-risk model was found to rely on data that is only available after a ride is cancelled or completed; once that leakage is removed, the model has no real predictive power, and a genuine early-warning system would need new features not currently captured in the Hive schema.

These findings — together with the data-quality notes in Section 2 — are intended to be read alongside the technical README, closing the gap between "how the system was built" and "what the data actually tells the business."